GRAINGUARD · FEASIBILITY ASSESSMENT

GrainGuard feasibility assessment

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	GGD-FEA-001	Version	0.1
Status	Draft	Date	2026-10-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/grainguard		
Source	docs/08-feasibility.md @ 5c553ac		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First issue (kit 1.8 upgrade)

GrainGuard feasibility assessment

GrainGuard is feasible on paper: the physics is sound and every part can be bought or made in a farm workshop, but two things only a trial can settle (whether interstitial RH predicts moisture to ±0.8 points, and what a thin cable really pulls on a bin's roof hanger) and one decision for Amish (the battery's disconnect setting against cold, GGD-DEC-001 open item 11) stand between it and a working prototype. Cost is the weakest aspect: USD 410.50 per bin against a USD 260 value-engineering target. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	The working principle is within known science (22 of 30 claims within, 7 plausible to confirm by measurement, 1 fix proposed); the decision rule's key input, the fan heat, is measured by the plenum probe; the D3 model of all 331 parts passes the fidelity and interference checks	High	Moisture estimate at risk (PR-002); a center cable watches 3.3 % of the cross-section (PR-006)	GGD-STM-001; GGD-CAL-001 B, C; `cad/fidelity.json`
Materials	Galvanized and stainless steel, aluminium sheet and angle, printed ASA, polycarbonate box; all common. ASA pods and coated boards face grain dust and phosphine	Medium	Phosphine resistance of the coated boards unverified (S28); ASA outdoors and in grain to confirm	GGD-STM-001 S28; BOM lines 2, 4, 9
Manufacturing and assembly	Cutting, drilling and bending angle and sheet, one small weldment (or a local fabricator), 3D printing and screw-terminal wiring; the build plan covers every part in order, and the D3 model fixed five fit details (D11)	High	Roof work and an empty-bin entry to fit the cable need a confined-space procedure and fall protection	GGD-BLD-001; GGD-DEC-001 D11
Power and controls	3.28 Wh per day; 7.7 days with no sun; 20 W panel refills in 2.9 winter days with the mast south of the bin (D13); fail-off relay, timeout and hand position	Medium	Thin refill margin (2.9 against 3 days); the battery's freezing point needs a higher disconnect setting, which cuts autonomy to 5.4 days (open item 11)	GGD-CAL-001 D, G; STEM check S13, S14, S17
Radio	16.9 dB margin at 1 km past one building; 24-byte packets within the 0.4 s dwell limit	High	Real farms add more obstructions; a site survey at TRL 4	GGD-CAL-001 E; GGD-REG-001 R1
Structure	Mast 2.7 times below yield with the stay; 1.7 times with the stay off in gusts under 30 m/s (D12); rope 4.3 times the largest force measured on any cable	Medium	Roof hanger rating per bin (PR-010); anchor capacity to confirm with the anchor maker (S26)	GGD-CAL-001 F
Supply of parts	Every line is a common retail item (sensors, LoRa module, AGM battery, PWM controller, 20 W panel, glands, galvanized pipe and angle); no single supplier needed	High	Prices are indicative, not quotes; the adjustable-disconnect controller and the labyrinth vents are estimates	`bom/bom.csv`; GGD-MKT-001
Cost	USD 410.50 per bin, USD 150.50 over the USD 260 value-engineering target; commercial cables and fan control cost several times more per bin	Low	The mast, pods and relay kit drive the cost; savings in GGD-DEC-001 (Value engineering) and the option C graft in GGD-OPT-001	GGD-CAL-001 H; GGD-DEC-001
Regulatory	Radio and battery ventilation met by design; the hazardous-location status of the pods inside a dusty bin is unassessed	Medium	To assess with a reviewer before any product claim	GGD-REG-001 A2

Ratings: High, Medium, Low.